UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 19

Fall 2025

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Convex relaxation is very important in *integer programming* (IP).

- LP subject to the additional constraint that all elements of $\boldsymbol{x}$ are integers is called IP.
- IP in standard form:

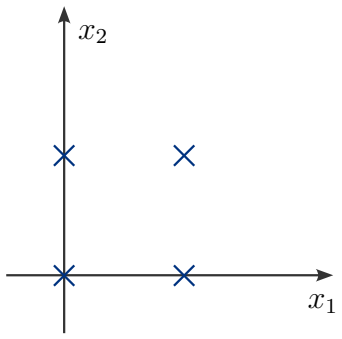
$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x}, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq \mathbf{0}, \\ & x_i \in \mathbb{Z},\ i = 1, \ldots, n. \end{aligned}$$

- IP is non-convex.

Example 1 (Integer programming)**.** Consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & x_1 - x_2, \\ \text{s.t.} \quad & x_1, x_2 \in \{0, 1\}. \end{aligned}$$

Feasible set:



Standard form:

$$\begin{aligned} \min_{x_1,x_2,s_1,s_2\in\mathbb{R}} \quad & \begin{bmatrix} 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix} \\ \text{s.t.} \quad & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \\ & \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix} \geq \mathbf{0}, \quad x_1, x_2, s_1, s_2 \in \mathbb{Z}. \end{aligned}$$

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Consider the following IP:

$$\begin{aligned} \min \quad & \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x}, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq \mathbf{0}, \\ & x_i \in \mathbb{Z}, \end{aligned} \tag{1}$$

and its convex relaxation:

$$\begin{aligned} \min \quad & \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x}, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq \mathbf{0}. \end{aligned} \tag{2}$$

- (2) is an LP that is a convex relaxation of IP (1).
- Recall that to find vertices of (2), we select m indices $i_1, \ldots, i_m$ and compute

  $$\begin{bmatrix} \boldsymbol{a}^{(i_1)} & \cdots & \boldsymbol{a}^{(i_m)} \end{bmatrix}^{-1} \boldsymbol{b}.$$

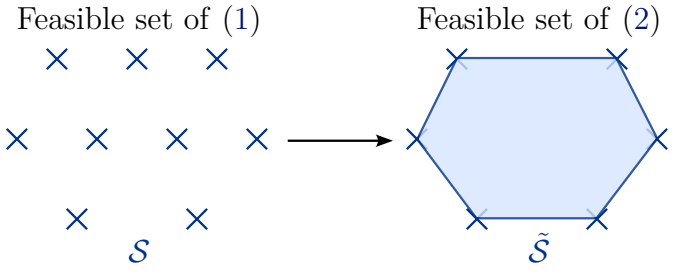
  This gives the non-zero entries of a vertex $\boldsymbol{y}$ (see Lecture Note 17).
- Therefore, all vertices of the feasible set of (2) are integral (meaning that its elements are all integers) iff, for every indices $i_1, \ldots, i_m$, if $\begin{bmatrix} \boldsymbol{a}^{(i_1)} & \cdots & \boldsymbol{a}^{(i_m)} \end{bmatrix}$ is invertible, then $\begin{bmatrix} \boldsymbol{a}^{(i_1)} & \cdots & \boldsymbol{a}^{(i_m)} \end{bmatrix}^{-1} \boldsymbol{b}$ is integral.

Theorem 1. *If all vertices of the feasible set of (2) are integral, then the convex relaxation is exact, meaning that:*

(i) The optimal objective value of (2) is equal to the optimal objective value of (1).

(ii) Find a vertex solution of (2) (note: (2) may have more solutions but one of them is a vertex) and that serves as a global minimum for (1).

Proof. Since (2) is a convex relaxation, we have $\mathcal{S} \subset \tilde{\mathcal{S}}$:

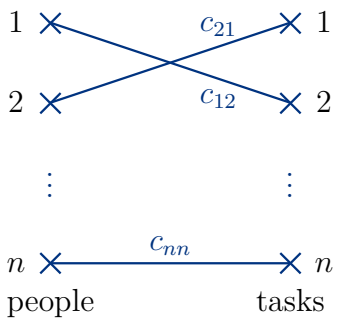


Since (2) is an LP, it has a solution that is a vertex. Let $\tilde{\boldsymbol{x}}$ denote such solution. By assumption, $\tilde{\boldsymbol{x}}$ is integral, thus $\tilde{\boldsymbol{x}} \in \mathcal{S}$. Since $\mathcal{S} \subseteq \tilde{\mathcal{S}}$, $\tilde{\boldsymbol{x}}$ is also a global minimum for (1). ■

For famous problems such as transportation and assignment, the vertices of (2) are indeed all integral:

Definition 1 (Assignment problem)**.** There are n people and n tasks. If person i works on task j, it takes c_{ij} minutes to complete. Assign each person to one task so that all tasks are assigned and the total time is minimized.

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Define variables x_{ij} for $i = 1, \dots, n$ and $j = 1, \dots, n$, where $x_{ij} = 1$ if person i works on task j and $x_{ij} = 0$ otherwise. We have the following IP:

$$\begin{aligned} \min \quad & \sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}x_{ij}, \\ \text{s.t.} \quad & \sum_{j=1}^{n} x_{ij} = 1, && \text{(each person is assigned to a task)} \\ & \sum_{i=1}^{n} x_{ij} = 1, && \text{(each task is assigned to a person)} \\ & x_{ij} \geq 0, \quad x_{ij} \in \mathbb{Z}. \end{aligned}$$

(We don't need to say $x_{ij} \in \{0, 1\}$ since the constraints $\sum x_{ij} = 1, x_{ij} \geq 0, x_{ij} \in \mathbb{Z}$ already guarantee that $x_{ij} \in \{0, 1\}$.)
LP relaxation:

$$\begin{aligned} \min \quad & \sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}x_{ij}, \\ \text{s.t.} \quad & \sum_{j=1}^{n} x_{ij} = 1, \ \forall i, \\ & \sum_{i=1}^{n} x_{ij} = 1, \ \forall j, \\ & x_{ij} \geq 0, \ \forall i, j. \end{aligned} \tag{3}$$

Theorem 2. *All vertices of* (3) *are integral, so the convex relaxation is exact.*

(Note: if we want to put (3) in standard form, then we should define $\boldsymbol{x} = [x_{11}, x_{12}, \dots, x_{nn}]^{\mathrm{T}}$ as the vector of variables which has n^2 elements.)
Assignment is a special case of transportation.

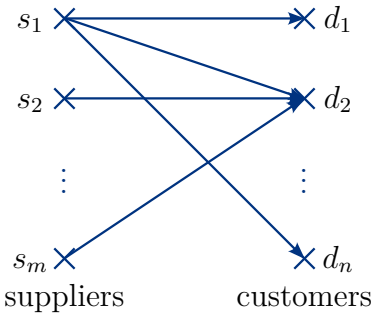
Definition 2 (Transportation problem)**.**

- Consider m suppliers selling a commodity (such as iPhone). Supplier $i \in \{1, \dots, m\}$ has s_i items.
- Consider n customers receiving the commodity. Customer $j \in \{1, \dots, n\}$ needs d_j items.

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- Define c_{ij} to be the cost of shipping an item from supplier i to customer j.
- Define x_{ij} to be the number of items sent from supplier i to customer j (unknown and to be optimized).

(Example: say customer 1 needs 10 items, then they can get 2 from supplier 1 and 8 from supplier 2.)



To minimize the total cost, we have the following IP:

$$\begin{aligned}
\min_{x_{ij}} \quad & \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}x_{ij}, \\
\text{s.t.} \quad & \sum_{j=1}^{n} x_{ij} \le s_i,\ i = 1,\dots,m, && \text{(supply constraint)} \\
& \sum_{i=1}^{m} x_{ij} = d_j,\ j = 1,\dots,n, && \text{(demand constraint)} \\
& x_{ij} \ge 0,\ x_{ij} \in \mathbb{Z},\ \forall i,j.
\end{aligned}$$

LP relaxation: drop the integral constraint $x_{ij} \in \mathbb{Z}$.

Theorem 3. *In Definition 2, assume $s_1,\dots,s_m,d_1,\dots,d_n$ are integers and $s_1+\cdots+s_m = d_1+\cdots+d_m$ (total supply equals total demand), then the LP relaxation is exact.*

If $m = n$, $s_1 = \cdots = s_m = d_1 = \cdots = d_n = 1$, then transportations reduces to assignment.

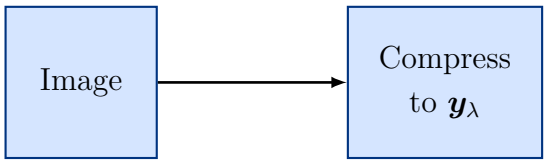
Applications of LP and QP in learning sparse vectors.

- $\boldsymbol{x} \in \mathbb{R}^n$ is called *sparse* if many of its entries are zero. Otherwise, it is called *dense*.
- The ratio of number of zeros to number of non-zero entries determines the level of sparsity.
- Example of sparse vector: $\boldsymbol{x} = [0,0,1,0,0,0,1]^{\mathrm{T}}$.

Example 2 (JPEG 2000 compression protocol)**.** Consider an arbitrary 256×256 grayscale image.

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- Each pixel has an integer value in the range $[0, 255]$ showing its color, where 0 indicates black and 255 indicates white.
- To store the image, we convert it to a vector $\boldsymbol{x} \in \mathbb{R}^{256^2} = \mathbb{R}^{65,536}$ (put all columns of the matrix on top of each other to make a tall vector).
- Let $\mathbf{A} \in \mathbb{R}^{65536 \times 65536}$ be a specific orthonormal matrix (obtained from wavelet transform or discrete Fourier transform).
- Define $\boldsymbol{y} = \mathbf{A}\boldsymbol{x}$. It turns out that most elements of $\boldsymbol{y}$ are close to zero for real-world pictures (this fact is used in image compression).
- Say $\boldsymbol{y} = [10, -0.0001, 0.0002, \dots]^{\mathrm{T}}$. Define $\boldsymbol{y}_\lambda$ to be a vector obtained from $\boldsymbol{y}$ by replacing each element in the interval $[-\lambda, \lambda]$ with 0. λ is chosen to be small.
- Now, $\boldsymbol{y}_\lambda$ is a sparse vector. Storing $\boldsymbol{y}_\lambda$ is much easier than $\boldsymbol{x}$ since we only need to store non-zero values and their locations (indices).



- Now, to recover $\boldsymbol{x}$ from $\boldsymbol{y}_\lambda$, since $\boldsymbol{y} = \mathbf{A}\boldsymbol{x}$, we should calculate $\mathbf{A}^{-1}\boldsymbol{y}_\lambda = \mathbf{A}^{\mathrm{T}}\boldsymbol{y}_\lambda$ (since $\mathbf{A}$ is orthonormal).
- Since $\boldsymbol{y} - \boldsymbol{y}_\lambda \approx \mathbf{0}$, when $\mathbf{A}^{\mathrm{T}}\boldsymbol{y}_\lambda$ is reshaped into a matrix and converted to an image, a little bit of quality of the image is lost (check Example 9.19 for pictures).

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- In image processing, $\mathbf{A}$ provides a sparse representation of an image. In many other applications, there is a similar transformation that gives us a sparse representation.
- This area is called *compressed sensing*.

Consider a system:

- Let $\boldsymbol{x}^* \in \mathbb{R}^n$ denote an underlying state for the system that is unknown.

$$\boldsymbol{x}^* = \text{vector of some unknown parameters of interest.}$$

- To learn $\boldsymbol{x}^*$, we take measurements.
- Each measurement is in the form:

$$b_i = \boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x}^*, \quad i = 1, \dots, m,$$

where b_i is the scalar measurement and $\boldsymbol{a}_i$ is a known vector modeling the measurement.

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- We have m measurements and n unknowns.
- If $m < n$, it is underdetermined.

  For example, let $m = 3$, $n = 6$:

$$\begin{aligned} 1 &= x_1^* - x_2^* + x_3^* + x_4^* + x_5^* + x_6^*, \\ 5 &= -2x_1^* + 0 + 0 + 5x_4^* - x_5^* - x_6^*, \\ 4 &= -x_1^* - x_2^* - x_3^* + x_4^* + 0 - 5x_6^*. \end{aligned}$$

  We can't determine $\boldsymbol{x}^*$ from these equations.

- What if $\boldsymbol{x}^*$ is sparse? Say, only two of its entries are non-zero. Then we can set 4 entries to zero and see if we can solve for the remaining entries via the equations:

$$\begin{bmatrix} ? \\ ? \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 \\ ? \\ 0 \\ ? \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ ? \\ ? \end{bmatrix} \quad \cdots$$

  The number of scenarios is exponential in general.

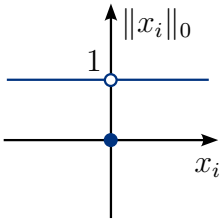
Let's formulate it as an optimization:

$$\min_{\boldsymbol{x} \in \mathbb{R}^n} \|\boldsymbol{x}\|_0 \ \text{ s.t. } \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \quad \left(\mathbf{A} = \begin{bmatrix} \boldsymbol{a}_1^{\mathrm{T}} \\ \vdots \\ \boldsymbol{a}_m^{\mathrm{T}} \end{bmatrix}, \ \boldsymbol{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \right)$$

where $\|\boldsymbol{x}\|_0$ is the cardinality of $\boldsymbol{x}$, i.e., the number of non-zero elements of $\boldsymbol{x}$.

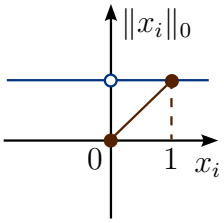
- The above optimization tries to find the sparsest vector $\boldsymbol{x}$ satisfying the measurements.
- The issue with the above optimization is that $\|\boldsymbol{x}\|_0$ is a non-convex function:

$$\|\boldsymbol{x}\|_0 = \|x_1\|_0 + \cdots + \|x_n\|_0, \quad \text{where } \|x_i\|_0 = \begin{cases} 0, & \text{if } x_i = 0, \\ 1, & \text{if } x_i \neq 0. \end{cases}$$

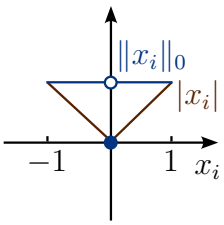


  For example, the segment between 0 and 1 is not above the function (thus non-convex):

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- Let's restrict x_i to $[-1, 1]$. Then, let's compare $\|x_i\|_0$ with $|x_i|$, which is convex:



Note that $|x_i| \leq \|x_i\|_0$, thus $\|\boldsymbol{x}\|_1 \leq \|\boldsymbol{x}\|_0$ if $\boldsymbol{x} \in \{\boldsymbol{y} \mid -1 \leq y_i \leq 1,\ i = 1, \ldots, n\}$.

In Lecture Note 18, we defined the concept of a *convex relaxation* for non-convex optimization problems with a convex objective function (such as IP). In that setting, since the objective function is already convex, the relaxation modifies only the feasible set while keeping the objective unchanged.
Here, we extend the definition of convex relaxation to optimization problems with *non-convex objective functions* (e.g., $\|\boldsymbol{x}\|_0$). In this case, we construct a convex relaxation by replacing the original non-convex objective with a convex lower bound, as formalized below.

Definition 3 (Convex relaxation)**.** Consider the two optimization problems:

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & \boldsymbol{x} \in \mathcal{S}, \end{aligned} \tag{4}$$

and

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \tilde{f}(\boldsymbol{x}) \\ \text{s.t.} \quad & \boldsymbol{x} \in \tilde{\mathcal{S}}, \end{aligned} \tag{5}$$

where the following conditions hold:

(1) $\tilde{f}(\boldsymbol{x})$ is convex;

(2) $\tilde{\mathcal{S}}$ is a convex set;

(3) $\mathcal{S} \subseteq \tilde{\mathcal{S}}$;

(4) $\tilde{f}(\boldsymbol{x}) \leq f(\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathcal{S}$.

Then, problem (5) is called a *convex relaxation* of problem (4). This generalizes the definition introduced in Lecture Note 18 to handle non-convex objectives.

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- Going back to the compressed sensing problem:

$$\begin{aligned} \min \quad & \|\boldsymbol{x}\|_0, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}. \end{aligned} \tag{6}$$

  We can replace the non-convex objective $\|\boldsymbol{x}\|_0$ with a convex lower bound:

$$\begin{aligned} \min \quad & \|\boldsymbol{x}\|_1, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \end{aligned} \tag{7}$$

- Note that (7) is a convex relaxation of (6) only when its feasible set is restricted to the box $\{\boldsymbol{y} \mid -1 \leq y_i \leq 1,\ i = 1, \ldots, n\}$.
- The solution to (7) may not be in the box, so (7) is not a relaxation of (6) in general.
- Let $\boldsymbol{x}^*$ be the solution of (7). There is a theorem in compressed sensing that says if
  (1) $\mathbf{A}$ is random enough rather than being structured and having many zero entries (e.g., elements of $\mathbf{A}$ come from a Gaussian distribution);
  (2) m (number of measurements) is greater than some constant times the number of non-zero elements in $\boldsymbol{x}^*$ times $\log n$;

  then $\boldsymbol{x}^*$ is the unique solution of (6).
- The exact statement of the theorem is beyond the scope of this class. But, it says a non-convex learning problem can be solved via a convex relaxation under some conditions.
- Note: (7) can be reformulated as LP:

$$\begin{aligned} & \min_{\boldsymbol{x} \in \mathbb{R}^n} \ \|\boldsymbol{x}\|_1 \ \text{ s.t. } \mathbf{A}\boldsymbol{x} = \boldsymbol{b} \\ \Leftrightarrow \ & \min_{\boldsymbol{x}, \boldsymbol{t} \in \mathbb{R}^n} \ t_1 + \cdots + t_n \ \text{ s.t. } \mathbf{A}\boldsymbol{x} = \boldsymbol{b},\ -t_i \leq x_i \leq t_i,\ i = 1, \ldots, n. \end{aligned}$$

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